

Project Design Documentation

Section 03: Automated Support Ticket Classifier

1. Executive Summary

This document outlines the architectural design and implementation of a high-performance, local Machine Learning classifier designed to categorize customer support tickets into five distinct categories: **Billing**, **Technical Issue**, **Feature Request**, **Complaint**, and **Other**.

86%

Evaluation Accuracy

52.6ms

Avg. Latency

< 500ms

SLA Requirement

2. Dataset Engineering

2.1 Creation Process

A synthetic dataset was engineered to simulate real-world customer interactions. The data was generated using LLM augmentation to ensure semantic diversity while maintaining categorical boundaries.

- **Total Records:** ~700 entries.
- **Labeling Strategy:** Manual verification of category assignments to minimize noise.
- **Data Split:** An 80/20 train-test split was utilized during the development phase, followed by a final 100-row "Hold-out" evaluation set to verify real-world performance.

3. Model Selection & Training

3.1 Architecture: DistilBERT

We selected `distilbert-base-uncased` as the backbone. DistilBERT provides a 40% reduction in size compared to BERT-base while retaining 97% of its performance, making it the optimal choice for a local CPU-bound environment with strict latency requirements.

3.2 Training Strategy (Hybrid Cloud/Local)

To optimize resources and manage time constraints, a hybrid approach was adopted:

1. **Remote Training:** Utilized **Google Colab with a T4 GPU** for high-speed fine-tuning.
2. **Environment:** Managed dependencies via `transformers[torch]` and `datasets`.
3. **Artifact Export:** The trained model weights, tokenizer configurations, and vocabulary files were zipped and transferred to the local production environment.

Note: Due to significant time constraints, hyperparameter tuning was focused on rapid convergence, resulting in a 76% training match, which generalized to 86% on the final evaluation set.

4. Technical Performance & Results

All results are archived in `section_03_classifier/Results/`.

4.1 Categorical Precision

Category	F1-Score	Observation
Feature Request	0.97	Excellent identification of "wishlist" intent.
Billing	0.91	Highly accurate detection of financial queries.

Complaint	0.77	Moderate overlap with Technical Issues due to sentiment noise.
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4.2 Latency Audit

The system was tested on a local CPU to ensure compliance with the 500ms constraint. While a "Cold Start" peak was observed (430ms), the steady-state performance remained consistently below 40ms.

5. Conclusion & Future Improvements

The current implementation successfully meets the project SLA of < 500ms while maintaining high accuracy. Future iterations would focus on multi-label classification to better distinguish between "Technical Issues" and "Complaints" where semantic overlap is high.